Requirements Brief
Greenie

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1 Who we are

We are a start-up company developing a web app that encourages sustainable habits through a gamified eco-action tracker. Students create an account, log their eco-friendly activities and compare their weekly points on a leaderboard, either among all students or among the people they follow.

2 Project scope

Project start	ASAP
Deadline	Fri 6 Nov 2026 (first milestone: Sat 31 Oct 2026)
Budget	100 USD in total, estimated effort 8 to 12 hours
Users	Students in Zurich, registered with email and password and a profile (nickname, name, date of birth, university)
Platform	Responsive web app, desktop and mobile layout
Mock-up	Figma mock-up (Section 11)
Success	A student can log an action in at most 3 taps; points, streak and the weekly leaderboards (everyone and followed students) update correctly (see Section 8); the code runs from the README and is in a repository we own; the site is deployed on a free hosting tier

3 Deliverables

Milestone 1

- **Registration & Login:** Account creation with email and password plus profile fields (nickname, first and last name, date of birth, university) and a consent checkbox; log in and log out on any device (US-01, US-13).
- **Home Status:** Total points, current streak and points earned today (US-02).
- **Action Logging & Point Calculation:** Form interface to log an action from a fixed list of 8 actions; the action's fixed points are added, respecting its daily limit (US-03).
- **Daily Streaks:** Automated tracking of consecutive daily user activity (US-04).
- **Action Configuration File:** Admin control over `actions.json` to define action types, point values and daily limits without code changes (US-08).

Milestone 2

- **Find Students & Profiles:** Search for another student by nickname and view their profile (nickname, university, total points, points this week, streak) (US-11).

- **Follow System:** Follow and unfollow other students (US-12).
- **Weekly Leaderboards:** Two rankings based on points from the current week (Monday to Sunday, Zurich time), one for everyone (top 20 plus own rank) and one for the user and the people they follow; points update immediately and the ranking resets weekly (US-06).
- **Personal History (Should):** Individual user log displaying the last 20 logged eco-actions (US-05).
- **Info Guide (Should):** Static "Which bin in Zürich?" page driven by `guide.json`. We supply the content (at least 10 entries) before Milestone 2 starts; you build the page and loading logic only (US-07).
- **Delete My Account (Should):** Removal of the user's account, profile and all related data (US-09).
- **Password Reset (Should):** Reset of a forgotten password through an emailed link (US-10).
- **CO_2 Estimate (Should):** Every action carries an estimated CO_2 saving, defined in `actions.json`. It is shown when logging, on the success screen and as a running total on Home (US-14).
- **Badges (Could):** Six badges for points and streak milestones, defined in `badges.json`, with a "My badges" screen and a badge row on the student profile (US-15). Built only if time and budget remain.

Handover and operations (throughout, final at Milestone 2)

- **GitHub Repository:** Shared code repository maintained from day one, complete with seed data and a comprehensive `README.md` detailing setup, execution, and instructions for editing `actions.json` and `guide.json`.
- **Live Deployment:** Accessible website URL on a free hosting tier (front end, back end, database and login), reachable until 15 Nov 2026.
- **AI Governance:** Milestone-by-milestone **AI-use declaration** documenting how AI tools were integrated into the development process.

4 Out of scope

Barcode scanning, chatbot, address-based guidance, maps, native apps, photo verification, push notifications, payments, email confirmation at sign-up, sign-in with Google or university accounts, two-factor login, private or blocked profiles, direct messaging.

5 Backlog and expectations

Your role. Build against the acceptance criteria in Section 8, test, deploy, hand over the repository and declare any AI use.

Backlog. The prioritised backlog is Section 6 together with the user stories in Section 8. We work in two sprints that match the two milestones.

Expectations. A short written update every two days, early notice if a date is at risk, and a working increment at each milestone. Changes are agreed in writing first.

6 Scope and priorities

Priority	Item
Must	Register an account: email, password, unique nickname, first and last name, date of birth, university
Must	Log in and log out, on any device
Must	Log an action from a fixed list of 8 actions with fixed points and a daily limit
Must	Total points, points this week and current daily streak
Must	Find a student by nickname and view their profile
Must	Follow and unfollow students
Must	Weekly leaderboards: everyone (top 20 plus own rank) and the people I follow
Must	Action list, points and daily limits editable in one config file, without code changes
Must	Runs from README; repository owned by us; deployed on a free tier
Should	Own log history (last 20 entries)
Should	Static "Which bin in Zürich?" guide, content supplied by us
Should	Delete my account and all my data
Should	Reset a forgotten password by email
Should	Estimated CO_2 saving per action, shown when logging, on the success screen and as a total on Home
Could	Six badges for points and streak milestones (built only if time and budget remain; not required for acceptance or payment)

7 Content we supply

7.1 Actions

Stored in `actions.json` with the fields `key`, `label`, `points`, `daily_limit`, `co2_g`. We supply this file. The values below are the initial content. `co2_g` is an indicative estimate of the CO_2 equivalent saved, in grams per logged action. We confirm the figures against public sources before the start of the second milestone; the supplier does not need to verify them.

Key	Label	Points	Daily limit	CO_2 (g)
`recycle_bottle`	Recycled a bottle (PET or glass)	10	5	40
`recycle_can`	Recycled a can	10	5	100
`recycle_paper`	Recycled paper or cardboard	10	3	30
`reusable_cup`	Used a reusable cup or bottle	10	3	25
`reusable_bag`	Used a reusable bag	5	3	20
`plant_meal`	Ate a plant-based meal	15	3	1500
`green_commute`	Took public transport, bike or walked instead of a taxi or car	15	2	1000
`second_hand`	Bought second-hand or repaired instead of buying new	20	1	5000

7.2 Guide content (Should)

Stored in `guide.json` with the fields `item`, `where`, `note`. We deliver at least 10 entries before the start of the second milestone. The supplier builds the page and the loading logic only and does not write or change the content.

7.3 Badges (Could)

Stored in `badges.json` with the fields `key`, `label`, `rule`, `threshold`. We supply this file. `rule` is one of `logs` (number of logged actions), `points` (total points) or `streak` (longest streak ever reached).

Key	Label	Rule	Threshold	Shown as
`first_action`	First action	`logs`	1	Log your first action
`points_100`	100 points	`points`	100	Reach 100 total points
`points_500`	500 points	`points`	500	Reach 500 total points
`points_1000`	1,000 points	`points`	1000	Reach 1,000 total points
`streak_7`	7-day streak	`streak`	7	Log 7 days in a row
`streak_30`	30-day streak	`streak`	30	Log 30 days in a row

8 User stories and acceptance criteria

Every acceptance criterion (AC) is one pass/fail check at milestone review.

US-01 Register an account (Must)

As a new student, I want to create an account and a profile, so that I can log actions, appear on the leaderboards and be found by other students.

AC1: The registration form has seven required fields: email, password (at least 8 characters), nickname (3–20 characters: letters, digits, underscore), first name and last name (1–50 characters each), date of birth (a valid date in the past; I must be at least 16 years old), and university (2–60 characters).

AC2: The email must be a valid address and unique, case-insensitive. If it is already registered, an error message appears and no account is created.

AC3: Nicknames are unique, case-insensitive. If taken, an error message appears next to the field and no account is created.

AC4: Every validation error is shown next to its field. No account is created until all fields are valid.

AC5: The form shows a short privacy text (supplied by us) and a consent checkbox. The "Create account" button stays disabled until the box is ticked.

AC6: The password field is masked. The password is never stored, logged or sent in plain text.

AC7: After success I am logged in and land on Home with 0 points and streak 0.

AC8: Email, first name, last name and date of birth are never shown to other users or on any leaderboard. Other users see only nickname, university, points and streak.

US-02 See my status (Must)

As a student, I want to see my points and streak, so that I stay motivated.

AC1: Home shows total points, current streak, and points earned today.
AC2: Values equal what is stored (the sum of my log entries).
AC3: Home has a prominent "Log an action" button.

US-03 Log an action (Must)

As a student, I want to log that I did something green, so that I earn points.

AC1: The Log screen lists all actions from `actions.json` with label and points.
AC2: Tapping an action adds exactly its points, once per tap.
AC3: The success screen shows "+N points", the new total and the streak.
AC4: Each action has a daily limit (Europe/Zürich calendar day). Beyond the limit the app shows "Daily limit reached for this action" and adds no points.
AC5: Logging works with at most 3 taps from Home.
AC6: Each action on the Log screen also shows its CO_2 estimate (US-14).

US-04 Daily streak (Must)

As a student, I want a streak, so that I am encouraged to log every day.

AC1: A "day" is a calendar day in Europe/Zürich, midnight to midnight.
AC2: The current streak is the number of consecutive days, ending today or yesterday, with at least one logged action.
AC3: If nothing was logged yesterday, the streak is 0 until I log again, then 1.

Test cases (log history and today's date → expected streak):

History	Today	Expected streak
Logged Mon, Tue, Wed	Wed	3
Logged Mon, Tue, Wed; nothing yet today	Thu	3 (until end of Thursday)
Logged Mon, Tue, Wed; nothing on Thu	Fri	0
As above, then logs on Fri	Fri	1
Two actions on the same day	any	counts as one day
Action at 23:59 and another at 00:01	next day	two consecutive days

US-05 My history (Should)

As a student, I want to see my recent logs, so that I can check what I did.

AC1: Lists my last 20 entries: action label, points, local date and time, newest first.
AC2: Empty state: "Nothing logged yet."

US-06 Weekly leaderboards (Must)

As a student, I want to see how I rank among everyone and among the people I follow, so that I can compare myself with others.

AC1: The Leaderboard screen has two tabs: "Everyone" and "Following".
AC2: Both tabs rank by points earned in the current week (Monday 00:00 to Sunday 23:59, Europe/Zürich). A new week starts at 0 points.
AC3: "Everyone" shows the top 20: rank, nickname, points. "Following" shows me and every student I follow, in the same columns.

AC4: My own row is highlighted. If I am outside the top 20 on "Everyone", my row and rank are shown below the list.
AC5: Users with equal points are ordered alphabetically by nickname and get consecutive ranks (no shared ranks).
AC6: The leaderboards reflect a new log within 5 seconds of a page refresh.
AC7: If I follow nobody, "Following" shows "You are not following anyone yet" and a "Find students" button.
AC8: Tapping a row opens that student's profile (US-11).

US-07 Zürich guide (Should)

As a student, I want to look up where an item goes, so that I recycle correctly.

AC1: The Guide page lists all entries from `guide.json`.
AC2: A text filter narrows the list as I type.
AC3: Changing `guide.json` changes the page without code changes.

US-08 Change the action list without code (Must)

As the product owner, I want to edit points and actions in one file, so that I do not need change requests.

AC1: Editing `actions.json` (label, points, limit) changes the app after a restart or redeploy, with no code changes.
AC2: The README explains how.

US-09 Delete my account (Should)

As a student, I want to delete my account, so that my personal data is removed when I leave.

AC1: A "Delete my account" button removes my login, profile data (email, name, date of birth, university, nickname), logs and follow relations after a confirmation.
AC2: I disappear from all leaderboards, search results and other students' following lists, and I can no longer log in.

US-10 Reset my password (Should)

As a student, I want to reset a forgotten password, so that I do not lose my account.

AC1: "Forgot password?" on the login screen asks for my email and always answers "If this email has an account, we sent a link", whether or not the account exists.
AC2: The emailed link lets me set a new password of at least 8 characters. The link expires (the platform's default is acceptable).
AC3: After a reset, the new password works and the old one no longer does.
AC4: The supplier states the free-tier email sending limits in the bid.

US-11 Find students and view profiles (Must)

As a student, I want to find other students and see their profile, so that I can decide whom to follow.

AC1: A search field finds students by nickname (partial match, case-insensitive) and shows up to 20 results with nickname and university.
AC2: If nothing matches, the app shows "No student found".

AC3: Tapping a result opens the profile: nickname, university, total points, points this week, current streak, and a Follow or Unfollow button.
AC4: A profile never shows email, first name, last name or date of birth.

US-12 Follow and unfollow (Must)

As a student, I want to follow other students, so that I have a leaderboard with the people I care about.

AC1: The button on a profile toggles between "Follow" and "Unfollow". The state is still correct after a page refresh.
AC2: Following is one-way: the other student does not need to approve and is not notified.
AC3: I cannot follow myself, and I cannot follow the same student twice.
AC4: A new follow appears on my "Following" leaderboard (US-06) within 5 seconds of a page refresh.

US-13 Log in and log out (Must)

As a registered student, I want to log in with my email and password on any device, so that I keep my points and follows.

AC1: The login screen has an email field, a masked password field, a "Log in" button, and links to registration and "Forgot password?".
AC2: Correct credentials open Home. Wrong credentials show one message, "Email or password is incorrect", without saying which part was wrong.
AC3: Login works on any device or browser, and shows the same data.
AC4: I stay logged in for at least 7 days or until I tap "Log out".
AC5: After "Log out", protected pages are no longer reachable: opening one redirects to the login screen.
AC6: A visitor who is not logged in sees only the login and registration screens.
AC7: Repeated failed logins are rate-limited (the platform's default is acceptable).

US-14 CO_2 estimate (Should)

As a student, I want to see roughly how much CO_2 my actions save, so that I understand their impact.

AC1: Each action in `actions.json` has a `co2_g` value: a whole number of grams, zero or more.
AC2: The Log screen shows the estimate next to each action, for example "$\approx$ 40 g CO_2". Values of 1000 g or more are shown in kg with at most one decimal.
AC3: The success screen shows the estimate of the action just logged.
AC4: Home shows my total, the sum over all my log entries, as "$\approx$ N CO_2 saved". A new user sees "$\approx$ 0 g CO_2 saved".
AC5: The estimate is stored with each log entry when it is logged. Later changes to `actions.json` do not change earlier entries.
AC6: Estimates are always shown with "$\approx$" and never as exact figures. Editing `co2_g` in `actions.json` changes the app after a restart or redeploy, with no code changes (as in US-08).

US-15 Badges (Could)

As a student, I want to earn badges for milestones, so that I have goals beyond the weekly ranking.

AC1: The six badges and their rules come from `badges.json` (Section 7.3).

AC2: A badge is earned when its rule is first met and stays earned. Rules are evaluated from my log entries; the streak rule uses my longest streak ever, computed as in US-04.
AC3: A "My badges" screen, reachable from Home, shows all six badges: earned ones in colour with name and rule, the others greyed out and marked "Locked". With no badge earned it shows "0 of 6 earned".
AC4: If a log earns a new badge, the success screen shows "New badge!" and the badge name.
AC5: A student's profile (US-11) shows the badges that student has earned, and no other personal data.
AC6: Badges never change points, streaks or rankings.

9 Non-functional requirements

Area	Requirement
Platform	Responsive web app, mobile-first (works from 360 px width), latest Chrome, Safari and Firefox. No native app.
Language	UI in English.
Time	All "days" use Europe/Zürich. Timestamps are stored in UTC.
Performance	Pages load in under 3 s on a normal mobile connection.
Data and privacy	Store only: email, password (as a salted hash, never in plain text), nickname, first name, last name, date of birth, university, follow relations, log entries. No third-party trackers. Email, first name, last name and date of birth are never shown to other users. The supplier works with seed data and test accounts only and keeps no copy of real user data after handover. Data is stored in an EU region. The free plan has no backups, so the seed data stays in the repository.
Hosting	Free tier only. Running cost 0 USD. The front end is served as static files from our repository (GitHub Pages is acceptable). Back end, database and login run on one free-tier managed service in an account owned by us, without a payment method; the supplier is added as a developer. The supplier names the service in the bid and states what happens after inactivity (pausing, expiry). The README explains how to keep it running and how to restore it. The deployed instance stays reachable until at least 15 Nov 2026. Preferred service: Supabase (free plan), created by us; the supplier may propose an alternative that meets all rules above and must justify it in the bid.
Technology	Front end: supplier's choice of mainstream technology; open source preferred. Back end: see Hosting. No paid services or licences. Stack named in the README.
Code structure	Front-end and back-end code in separate files or folders (no markup mixed into server logic).
Code quality	Block-level comments explaining each module and function. Clear variable names.
Security basics	Validate input, parameterised queries, no secrets in the repository. Use the platform's built-in authentication or a standard salted hash (bcrypt or argon2). HTTPS only.
Accessibility	Readable contrast, labelled buttons and fields.
Generative AI	AI coding assistants allowed only if declared in writing per milestone (tools and purpose). No AI-written content in `guide.json`, `actions.json` or `badges.json`.

10 Data model

Table or file	Fields
`users`	id, email (unique, case-insensitive), password_hash, nickname (unique, case-insensitive), first_name, last_name, date_of_birth, university, created_at
`follows`	follower_id, followed_id, created_at (each pair at most once; follower and followed must differ)
`action_logs`	id, user_id, action_key, points, co2_g, logged_at_utc, local_date_zurich
`actions.json`	key, label, points, daily_limit, co2_g
`guide.json`	item, where, note
`badges.json`	key, label, rule (logs, points or streak), threshold

Derived values: total points is the sum of `points` per user. Points this week are the sum of `points` with `local_date_zurich` in the current Monday-to-Sunday week. The streak is computed from `local_date_zurich` as in US-04. Total CO_2 is the sum of `co2_g` per user. Earned badges are computed from the log entries using the rules in `badges.json` and need no table of their own. The "Following" leaderboard ranks the user plus all `followed_id` values of that user. Email and password hash may live in the platform's authentication service; `users` then references that account.

11 Screens (Figma mock-up)

A view-only Figma mock-up accompanies this brief: [Figma link]. It has two pages: "Mobile" (390×844) and "Desktop" (1280×900) with the same frames. Each frame is named `S1 ...`, `S2 ...` and a label above it lists the story IDs. On mobile, the bottom tab bar on S2, S3, S5 and S6 has: Home, Log, Leaderboard, Guide. On desktop, the same four entries form the top navigation bar. The pixel-art style (palette, pixel font, icons) is a visual reference. The supplier may redraw the icons and illustrations in their own pixel style.

Frame	Elements	States shown	Stories
S0 Log in	App name, email field, masked password field, "Log in" button, links "Create account" and "Forgot password?"	Default; error "Email or password is incorrect"	US-13

Frame	Elements	States shown	Stories
S1 Register	App name, one-line pitch, fields (email, password, nickname, first name, last name, date of birth, university), privacy text with consent checkbox, "Create account" button	Default; errors "Email already registered", "Nickname taken", "Password needs at least 8 characters", "3–20 characters, letters, digits, _", "You must be at least 16"; button disabled until consent ticked	US-01
S2 Home	Total points (large), streak with flame icon, "Today: +N", "≈ N CO_2 saved", button "Log an action", entries "My history", "My badges" (Could) and "Find students", link "Log out"	New user (0 and 0); active user	US-02, 04, 05, 11, 13, 14, 15
S3 Log action	Title, list of 8 actions, each with label, "+N" and "≈ N g CO_2"	Default; limit-reached message on one action	US-03, 14
S4 Success	"+N points", new total, streak, "≈ N g CO_2 saved", buttons "Back to Home" and "Log another"	Streak increased; new badge (Could)	US-03, 04, 14, 15
S5 Leaderboard	Tabs "Everyone" and "Following"; rows: rank, nickname, points; own row highlighted; own rank row pinned below on "Everyone"	Default; user outside top 20; empty list; "Following" empty with "Find students" button	US-06

Frame	Elements	States shown	Stories
S6 Guide	Text filter, list of entries (item, where, note)	Default; no results	US-07
S7 History (Should)	List: label, points, date and time	Default; empty	US-05
S8 Find students	Search field, result list (nickname, university)	Default; results; "No student found"	US-11
S9 Student profile	Nickname, university, total points, points this week, streak, earned badges (Could), button "Follow" or "Unfollow"	Not following; following	US-11, 12, 15
S10 Reset password (Should)	Email field, "Send link" button; new-password form reached from the emailed link	Default; confirmation message; new password saved	US-10
S11 My badges (Could)	Six badge tiles with name and rule; earned in colour, others greyed with "Locked"	Some earned; none yet ("0 of 6 earned")	US-15

Navigation: S0↔S1; S0→S10; S0, S1→S2 after login or registration; S2→S0 on "Log out"; S2→S3→S4→S2; tab bar between S2, S3, S5, S6; S2→S7; S2→S8; S2→S11; S5 ("Following" empty state)→S8; S8→S9; S5 row→S9.

12 Delivery and acceptance

Commercial terms. Fixed price, paid on the platform in two equal milestones. The total price is agreed in the contract and stays within the budget stated in the job post.

Milestone	Content	Target date	Payment
M1	US-01, 02, 03, 04, 08, 13; repository; deployed skeleton	Sat 31 Oct 2026	50% of price
M2	US-06, 11, 12; Should: US-05, 07, 09, 10, 14; Could (optional, not required for payment): US-15; final fixes; handover	Fri 6 Nov 2026	50% of price

Acceptance procedure. We test every acceptance criterion on the deployed URL within 24 hours of delivery and send a defect list. The supplier fixes defects within 24 hours at no extra charge. Payment is released on acceptance, on the platform only.

Change requests. In writing, with price and date agreed before work starts.

Escalation. No reply within 24 hours triggers a reminder. No reply within 48 hours means we may end the contract and keep the paid-for work.